A sharp threshold for three-colour targets from four-colour cubes

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Abstract

Any collection of 29 cubes, each displaying all four colours of a fixed palette, contains eight that form a $2 \times 2 \times 2$ cube with monochromatic faces using exactly three colours. For this fixed palette, exactly two collections of 28 cubes fail, and reflection exchanges them. Only 28 of the 120 target varieties are needed for the guarantee. This answers the threshold question in Problem 2 of Berkove, Eickemeyer and Schulman (2018). The proof uses their corner-containment reduction and elementary counting. The threshold and classification are formalised in Lean.

1 Introduction

Berkove, Eickemeyer and Schulman (BES) [1, Problem 2] ask for the least number of four-colour cubes that guarantees a choice of eight cubes forming a three-colour corner solution: a $2 \times 2 \times 2$ cube whose exterior faces are monochromatic and use exactly three colours. Here every source cube uses all four colours of a fixed palette. A *variety* is a face colouring up to proper rotation. Collections are multisets of varieties, so repeated copies are allowed.

We prove that the least number is 29 and classify all collections of 28 cubes that fail. There are exactly two such collections, exchanged by reflection and by any odd colour permutation, so there is one obstruction up to colour permutation. Each consists of seven copies of each of four varieties, one with a monochromatic corner of each colour. The four varieties share the same corners with three distinct colours: they belong to one orientation class. Figure 1 shows one collection, and Theorem 3 gives its face tuples.

The classification explains the form $28 = 4 \cdot 7$. Mixing orientation classes allows new corner matches, while each colour bucket holds at most seven cubes without yielding a solution. Together these restrictions force an extremal collection into one orientation class, then force the four monochromatic-corner varieties to occur equally often. We first prove this rigidity, deduce the upper bound, and then verify that both obstructions fail.

2 Model and geometric facts

Reflections are not allowed as moves. Only exterior faces must match the target; internal faces are unrestricted.

A corner type is a cyclically ordered triple of face colours, read counterclockwise from outside along the corner's body diagonal. Equivalently, order the three outward face normals to form a positively oriented basis. A trichromatic corner has three distinct colours and two orientations, *abc* and *acb*. BES [1, Section 2] use clockwise order; our convention reverses every triple, preserving containment and matching. A source cube fits a target corner by a proper rotation if and only if it has the same cyclic corner type: the cyclic alignments of two corner frames are exactly those realised by proper rotations of the cube. A target is represented by its multiset of eight corner types: a collection builds it exactly when these eight occurrences can be matched to distinct cubes having the required types.

For distinct colours a, b, c, write $Q(a; b, c)$ for a target whose b and c faces are opposite and whose other four faces have colour a. Its corner multiset is $4aab + 4aac$. Write $R(a; b, c)$ for the analogous target with b and c adjacent; its corners are

$$2aaa + 2aab + 2aac + abc + acb.$$

Finally, write $D(a, b, c)$ for the target with opposite pairs aa, bb, cc; it has four corners of each orientation on $\{a, b, c\}$. We need only these three families of exactly-three-colour targets for the sufficient constructions.

If every corner type of a variety v occurs on a variety w, replacing a copy of w by v cannot create a solution to any target. In particular, it preserves unsolvability. This is the reduction of [1, Section 2], recorded in Table 1. Face tuples use the order $+x, -x, +y, -y, +z, -z$. The rows represent the seven orbits under colour permutations, with rotations already identified. They cover all 68 varieties, unlike the seven minimal representatives in [1, Figure 2]; four of the orbits are retained.

Representative	Orbit size	Retained type	Replacement
001123	6	checkerboard C	—
010203	8	triple-corner T	—
010213	12	pair type P	—
010223	12	pair type P	—
000123	12	—	001123
001213	12	—	001123
010123	6	—	001123

Table 1: The 68 varieties reduce to 38.

The retained types are precisely the minimal varieties in [1, Table 8]. A reduced variety belongs to bucket a if it has at least two of the three corner types aab with $b \neq a$. These are the buckets of [1, Definition 3], with each bucket's seven mirror pairs treated as fourteen individual varieties. The following lemma collects the geometric facts used in the proof. The orientation classes and checkerboard terminology are from [1, Section 2].

Lemma 1 (Finite geometry)**.** *Every variety can be reduced by corner containment to one of the 38 retained varieties in Table 1. Every non-retained variety can be replaced by a checkerboard. The retained varieties have these bucket incidences:*

- *There are eight T varieties. Each has a repeated colour a, corners aaa and aab for every $b \neq a$, and belongs only to bucket a.*
- *The 24 P varieties belong to two buckets, one for each repeated colour.*
- *The six checkerboards belong to no bucket. Their repeated colours occupy opposite face pairs; with repeated colours a, b, a checkerboard has both orientations on $\{a, b, c\}$ and $\{a, b, d\}$.*

The 32 ordinary varieties (T and P) split into two classes of 16, corresponding to BES's K_o and K_m (up to interchanging the names). The members of each class share four trichromatic corner types, one for each three-element colour set. The two classes have opposite orientations. In our convention one class has the types

$$\Omega = \{012, 031, 023, 132\},$$

one set of BES's universal corners*. Each class contains exactly one T variety for each repeated colour. Even colour permutations preserve the classes; odd colour permutations and reflection exchange them.*

Proof. The face-colour multiplicities are $3+1+1+1$ or $2+2+1+1$. In the first case, the three equal faces either meet at a vertex or contain an opposite pair, giving two orbits. In the second case, both repeated pairs can be opposite, exactly one can be opposite, or both can be adjacent. Fixing one adjacent pair and placing the other among the four remaining faces gives three patterns in the last case: 010213, 010223 and 010123. The second multiplicity case thus gives five orbits, completing Table 1. The three non-retained representatives contain the corner types of 001123; colour permutations transport these containments to their whole orbits. Table 2 gives the corner sets of the retained representatives. All three ordinary representatives in Table 2 belong to the class of Ω. Even colour permutations preserve Ω and odd ones reverse it, with triples read cyclically. Applying colour permutations to Table 2 gives the stated incidences and orientation classes. The eight T varieties split evenly between the classes, and even permutations act transitively on the four colours, so each class contains exactly one T variety for each repeated colour. Reflection reverses every trichromatic corner. □

Representative	Corner-type set
001123	$\{012, 013, 021, 031\}$
010203	$\{000, 001, 002, 003, 012, 023, 031, 132\}$
010213	$\{001, 003, 011, 012, 023, 031, 112, 132\}$
010223	$\{002, 003, 012, 022, 023, 031, 122, 132\}$

Table 2: Corner-type sets of the retained representatives, with cyclic triples written in lexicographically least form.

Lemma 2 (Bucket bound)**.** *Eight reduced cubes in bucket a can build a target $Q(a; b, c)$ for some b, c. Consequently, a reduced collection with no Q target has at most seven cubes in each bucket.*

Proof. Each of the eight cubes misses at most one of the three types aab. Thus at most one type is missed by five or more cubes. Choose two types, aab and aac, each missed by at most four cubes. Assign cubes missing aac to aab positions, and cubes missing aab to aac positions. These groups are disjoint and each has size at most four. Every remaining cube has both types, so they fill the remaining positions of $Q(a; b, c)$. □

3 The extremal collections

Theorem 1. *A collection of 28 four-colour cubes without a three-colour corner solution consists of seven copies of each of the four T varieties in one ordinary orientation class.*

Proof. The argument needs only the assumption that no target of type Q, R or D can be built. Replace each non-retained cube by a checkerboard using Lemma 1, leaving retained cubes unchanged. This preserves avoidance of all three target families. Let $A \geq B$ be the sizes of the two ordinary orientation classes, and let C count checkerboards, so $A + B + C = 28$. Call the class of size A the majority class (either one if equal). Call the remaining $M = B + C$ cubes *minority cubes*. A minority cube *supports* a three-element colour set E if it has the orientation on E opposite to the majority class. Smaller-class cubes support every E; a checkerboard supports two.

For $E = \{a, b, c\}$, let z_E count checkerboards supporting E. If $z_E \geq 8$, those eight checkerboards build the target $D(a, b, c)$. Thus $z_E \leq 7$ for each E, and, since each checkerboard supports two triples, $2C \leq 28$. It follows that $A + B \geq 14$, so $A \geq 7$. There are $m_E = B + z_E$ minority cubes supporting E. If $m_E \geq 4$, four such cubes and four majority cubes build $D(a, b, c)$. Hence $m_E \leq 3$ for every E. Summing gives

$$4B + 2C \leq 12, \qquad M \leq 6, \qquad A = 28 - M \geq 22. \tag{1}$$

From now on, let t_a and p_a count only majority-class T and P cubes in bucket a, and let n_T, n_P be their respective totals, so $A = n_T + n_P$. By Lemma 2, $t_a + p_a \le 7$. Each T cube belongs to one bucket and each P cube to two, so summing gives

$$n_T + 2n_P \le 28, \qquad n_P \le M. \tag{2}$$

We will repeatedly use the following construction. If six majority cubes supply $2aaa + 2aab + 2aac$ and a minority cube supports $\{a, b, c\}$, then the minority cube supplies the opposite trichromatic orientation and an unused majority cube supplies the other. Such a cube is available since $A \ge 22$. This builds $R(a; b, c)$.

Suppose $M > 0$. Every minority cube supports some triple containing any given colour a. The construction above therefore rules out $t_a \ge 6$. Thus $t_a \le 5$ for every a, and $n_T \le 20$. Since $n_T + n_P + M = 28$ and $n_P \le M$, we obtain $M \ge 4$.

For any two distinct colours a, b, some minority cube supports a triple containing both. If not, every minority cube would be a checkerboard. A checkerboard whose repeated pair meets $\{a, b\}$ supports such a triple, so all would have the complementary repeated pair $\{c, d\}$, supporting exactly $\{a, c, d\}$ and $\{b, c, d\}$. Thus $m_{\{a,c,d\}} = M \ge 4$, contradicting $m_E \le 3$.

We claim that $t_a \ge 4$ forces $t_a + p_a \le 5$. Otherwise, since $t_a \le 5$, choose six majority cubes in bucket a, with four or five T cubes and the rest P cubes. These one or two P cubes share a type aab, since each has two of the three possibilities. Choose a minority cube supporting $\{a, b, c\}$. The P cubes take aab positions and the T cubes supply the rest of $2aaa + 2aab + 2aac$, so the construction above gives a solution, proving the claim.

If $t_a \le 3$, the bucket bound gives $2t_a + p_a \le t_a + 7 \le 10$. If $4 \le t_a \le 5$, the claim gives $2t_a + p_a \le t_a + 5 \le 10$. Thus $2t_a + p_a \le 10$ in all cases. Each majority T cube contributes two through its single bucket, and each majority P cube contributes one through each of its two buckets. Summing gives

$$2A = 2n_T + 2n_P = \sum_a (2t_a + p_a) \le 40,$$

contradicting $A \ge 22$ from (1). Therefore $M = 0$, and (2) forces $n_P = 0$. All 28 cubes are majority-class T cubes, and the four bucket bounds force each $t_a = 7$.

Since the reduced collection contains no checkerboards, no replacement occurred: the original collection was already reduced. □

Theorem 2. *Every collection of 29 four-colour cubes contains a three-colour corner solution.*

Proof. Otherwise every 28-cube subcollection would have the form in Theorem 1. Take one such subcollection S and let v be the remaining cube. Remove from S a cube of a variety different from v, and insert v. The multiplicity of v's variety is now 1 or 8, whereas every variety in a collection described by Theorem 1 has multiplicity 0 or 7, a contradiction. □

4 Sharpness

Lemma 3. *In a cube using exactly three colours, the two orientations of a trichromatic corner occur equally often. If no trichromatic corner occurs, the cube is $Q(a; b, c)$ for some distinct a, b, c.*

Proof. For any cube coloured from a palette $\{a, b, c, d\}$, let $n(xyz)$ count corners of cyclic type xyz. Read the three directed transitions around each corner in the chosen cyclic order. Each edge between an a face and a b face contributes ab at one endpoint and ba at the other, so the total counts of these transitions agree. Corners with repeated colours contribute equally to both counts. The remaining contributions give

$$n(abc) - n(acb) + n(abd) - n(adb) = 0. \tag{3}$$

If only a, b, c occur, the last two terms vanish, proving the balance.

For the second assertion, if a colour c occurs on only one face, its four neighbours must all have the same colour a: a change from a to b around this face would produce a trichromatic corner. The opposite face is then the sole b face. Otherwise each colour occurs twice. If the two c faces are opposite, the four neighbours of either contain both a and b, forcing a change. If they are adjacent, the other three neighbours form a path containing both a and b, since each occurs only twice. Again a change is forced. Both cases produce a trichromatic corner. □

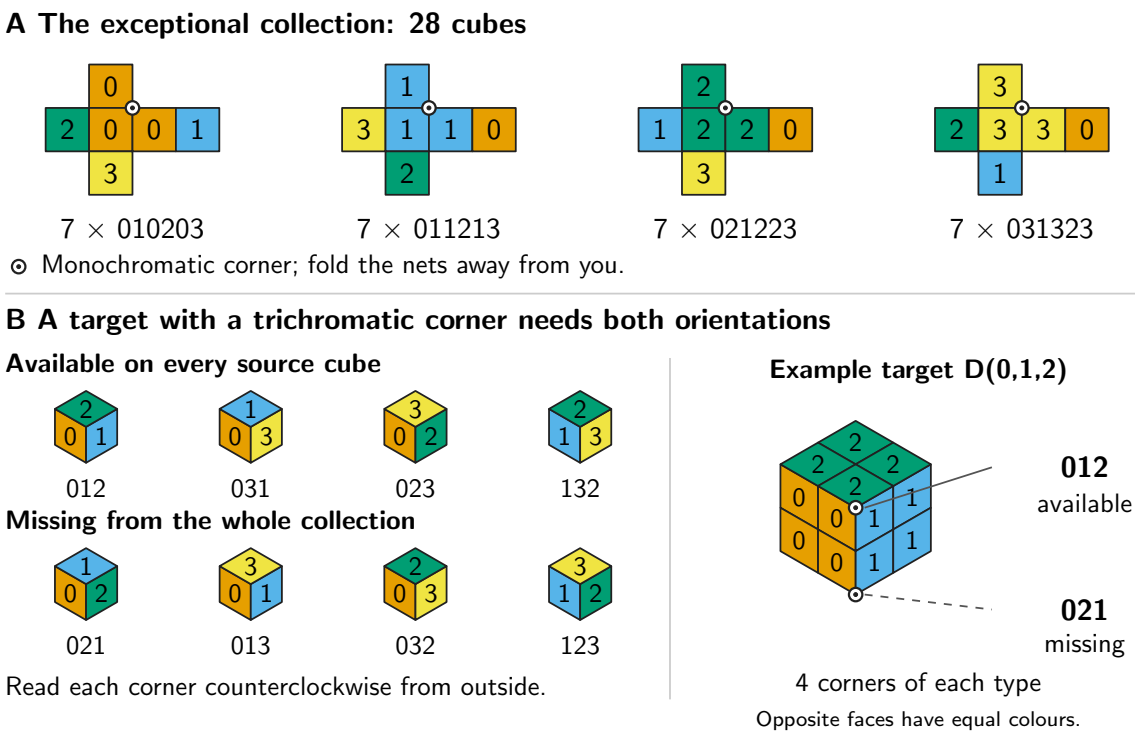


Figure 1: Why one exceptional collection fails. (A) Seven copies of each variety, with monochromatic corners marked. (B) Targets with a trichromatic corner need both orientations; the example's bottom face is hidden. (C) Targets without trichromatic corners require eight cubes with the same repeated colour, but only seven are available; a, b, c are distinct. Reflection gives the other obstruction.

Theorem 3. *The threshold 29 in Theorem 2 is best possible. For a fixed palette, exactly two 28-cube collections fail, and they are mirror images. Up to colour permutation, there is a single obstruction.*

Proof. Take seven copies of each of the following four face tuples (Figure 1):

$$010203, \qquad 011213, \qquad 021223, \qquad 031323.$$

These are T cubes with repeated colours $0, 1, 2, 3$, respectively, all in the same ordinary orientation class. A target with any trichromatic corner requires both orientations by Lemma 3, whereas the collection supplies only one. A target without one is $Q(a; b, c)$; every one of its eight corners has colour a twice, so only the seven T cubes of repeated colour a are usable. No three-colour target is possible. Reflection gives the analogous example in the other orientation class. Theorem 1 excludes all other collections. Every odd colour permutation exchanges the two orientation classes and hence these two collections. □

Corollary 1 (Restricted targets)**.** *Every collection of 29 four-colour cubes, and every solvable collection of 28, builds a target of type Q, R or D. These families contain just 28 of the 120 target varieties.*[1]

Proof. The classification argument uses only these three families, so a 28-cube collection avoiding them has one of the two extremal forms and is therefore unsolvable by Theorem 3. The deletion-and-insertion argument of Theorem 2 likewise rules out a 29-cube collection avoiding them. There are $4\binom{3}{2} = 12$ varieties each of Q and R: choose a and the unordered pair $\{b, c\}$. There are $\binom{4}{3} = 4$ varieties of D, one for each three-colour set. □

Remark. For the palette $\{0, 1, 2, 3\}$, write $\delta(xyz) = n(xyz) - n(xzy)$. Applying (3) to the different colour pairs gives

$$\delta(012) = \delta(031) = \delta(023) = \delta(132).$$

This common integer is the cube's *degree*, relative to Ω. Reflection negates the degree, as does each transposition of colour labels; thus even colour permutations preserve it and odd ones negate it. Each ordinary representative in Table 2 has eight distinct corner types, so each occurs once; the two ordinary classes therefore have degrees $+1$ and -1. The checkerboard representative has neither 023 nor 032, so its degree, and that of every checkerboard, is zero.

5 Formal verification and computation

The threshold and classification are formalised in Lean 4.35.0 [3] with mathlib v4.35.0 [4], for both inventories of the 68 source varieties and arbitrary indexed families of four-colour cubes. A solution selects eight distinct copies and proper rotations matching the exposed faces of a three-colour target. The threshold and sharpness also have statements without source or target catalogues; the classification counts source cubes in each rotation class. Corollary 1 is also formalised for arbitrary indexed families, using geometric definitions of Q, R and D.

Kernel checks establish catalogue coverage, source and target uniqueness up to rotation, the restricted-target counts, and correctness of stored corner codes. A palette-independent lemma, allowing repeated colours, equates cyclic corner matching with physical face matching by a proper rotation. Twenty-five results are restated in a standalone challenge file importing only mathlib. Their proofs were replayed through the Lean, NanoDa and con-ron kernels.

[1] For any fixed three colours, a rotation with k face cycles fixes $3^k - 3 \cdot 2^k + 3$ colourings using all three. The 24 rotations have $6, 4, 3, 2$ face cycles with multiplicities $1, 3, 12, 8$, so Burnside's lemma gives $(540 + 3 \cdot 36 + 12 \cdot 6)/24 = 30$ varieties. Four choices of omitted colour give 120.

The Lean proof uses counting and finite Hall certificates for the sufficient constructions. It treats checkerboards and the smaller ordinary class separately; the written proof combines them as minority cubes. The lower bound checks Hall deficiencies for all 120 targets and both extremal collections. Neither proof enumerates all collections.

The axiom audit finds only `propext`, `Classical.choice` and `Quot.sound`, all standard in Lean. No proof uses holes or `native_decide`. The repository[2] contains sources and reproduction commands. The cube-rotation and exposed-face definitions are reused from the Lean project accompanying [5].

A computational search found the 28-cube example and returned results consistent with the threshold and classification. A separate script checks both geometric tables, the stated incidences and the lower obstructions. Neither proof depends on these computations.

Literature and priority

A literature search on November 1, 2026 found no earlier answer; this does not establish priority. Rosenfeld's October 2026 article [2] *MacMahon Cubes and Moshe's Insanity, Puzzle 1* remains unread, and its relevance is undetermined.

Declaration of AI-generated research and review

GPT-6-Astra (OpenAI), used through Codex, developed the original research argument, computational search, Lean formalisation and manuscript under the author's direction. Claude Opus 5.5 (Anthropic), used through Claude Code, contributed the minority-cube argument, balance proof, further mathematical simplifications, illustrations and review. Subsequent Codex revisions incorporated and extended these contributions. Detailed attribution and review scope are recorded in the repository's `formalization.yaml`. The human author initiated and directed the research and accepts responsibility for the paper's content and the accuracy of this disclosure.

References

[1] E. Berkove, A. Eickemeyer and M. Schulman, *The 4-Color Cubes Puzzle*, Integers **18** (2018), A68. Journal PDF.

[2] M. Rosenfeld, *MacMahon Cubes and Moshe's Insanity, Puzzle 1*, Geombinatorics Quarterly **36**(2) (October 2026), starts at p. 78. Journal listing; full text not examined.

[3] L. de Moura and S. Ullrich, *The Lean 4 Theorem Prover and Programming Language*, in Automated Deduction—CADE 28, LNCS **12699**, Springer (2021), 625–635. doi:10.1007/978-3-030-79876-5_37.

[4] The mathlib Community, *The Lean Mathematical Library*, in Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM (2020), 367–381. doi:10.1145/3372885.3373824. arXiv:1910.09336.

[5] J. Davies, *The ten minimum universal sets of MacMahon cubes*, manuscript and Lean project (October 31, 2026). Repository.

[2] https://github.com/jasondavies/macmahon-four-to-three